Ex.No:06	JAVASCRIPT – TIMER FUNCTIONS AND ARRAYS

Objective:

To create web pages using various timer functions, events and objects in JavaScript.

Description:

Timing Events

The window object allows execution of code at specified time intervals. These time intervals are called timing events. The three key methods to use with JavaScript are:

- setTimeout(*function, milliseconds*)
  Executes a function, after waiting a specified number of milliseconds.
  Ex: window.setTimeout(*function, milliseconds*);
- setInterval(*function, milliseconds*)
  Same as setTimeout(), but repeats the execution of the function continuously.
  Ex: window.setInterval(*function, milliseconds*);
- The clearInterval() method stops the executions of the function specified in the setInterval() method.
  EX: window.clearInterval(*timerVariable*)

Date Object

Date Object gives various methods to multiple ways to access date, month, year details.

Method	**Description**
getDate()	Returns the day of the month (from 1-31)
getDay()	Returns the day of the week (from 0-6)
getFullYear()	Returns the year
getHours()	Returns the hour (from 0-23)
getMilliseconds()	Returns the milliseconds (from 0-999)
getMinutes()	Returns the minutes (from 0-59)
getMonth()	Returns the month (from 0-11)
getSeconds()	Returns the seconds (from 0-59)

getTime()	Returns the number of milliseconds since midnight Jan 1 1970, and a specified date
getDate()	Returns the day of the month (from 1-31)
getDay()	Returns the day of the week (from 0-6)
getFullYear()	Returns the year
getHours()	Returns the hour (from 0-23)
getMilliseconds()	Returns the milliseconds (from 0-999)
getMinutes()	Returns the minutes (from 0-59)
getMonth()	Returns the month (from 0-11)
getSeconds()	Returns the seconds (from 0-59)
getTime()	Returns the number of milliseconds since midnight Jan 1 1970, and a specified date
setTime()	Sets a date to a specified number of milliseconds after/before January 1, 1970
setDate()	Sets the day of the month of a date object
setFullYear()	Sets the year of a date object
setHours()	Sets the hour of a date object
setMilliseconds()	Sets the milliseconds of a date object
setMinutes()	Set the minutes of a date object
setMonth()	Sets the month of a date object
setSeconds()	Sets the seconds of a date object
toString()	Converts a Date object to a string
toTimeString()	Converts the time portion of a Date object to a string

Class and Object

A class is a type of function, but instead of using the keyword function to initiate it, we use the keyword class, and the properties is assigned inside a constructor() method.

Example:

```
class Car {
  constructor(brand) {
    this.carname = brand;
```

```
  }
  present() {
    return "I have a " + this.carname;
  }
}
mycar = new Car("Ford");
document.getElementById("demo").innerHTML = mycar.present();
```

Sample Questions:

1. Write JavaScript code to display digital clock.

```
<h1>Digital Clock!</h1>
<span id="time"></span>.
<script>
//define function display
//call the time display function for every one second
</script>
```

```
// get time
        var d= new Date();
// extract  hour,minutes and seconds
        d.getHours
        d.getMinutes()
        d.getSeconds()
// call the time display function for every one second
        setInterval("display()", 1000);
```

2. Write JavaScript code to change background colour for every 30 seconds.

```
<body>
<h1>Dynamic Background!</h1>
<script>
    //define function setBackgroundColor
    //call the setBackgroundColor function for every two seconds
</script>
</body>
```

```
  //create random colors using rgb function
     var red = Math.floor(Math.random()*255);
     var blue = Math.floor(Math.random()*255);
     var green = Math.floor(Math.random()*255);
      var color= "rgb("+red+","+green+","+blue+")";
  //change or set body background color
     document.getElementsByTagName("body")[0].style="background-
color:"+color;
```

3. Create event handlers that will increase the font size when a mouse moves over a paragraph and decrease the size when the mouse moves off the paragraph.

```
<h1>Change font size!</h1>
<p id="para1" onmouseover="incSize()" onmouseout="decSize()">
move cursor over me to zoom in...
</p>
<script>
    //define function incSize
    //define function decSize
</script>
</body>
```

```
// To change font size
function incSize(){
   document.getElementById("para1").style="font-size:28px";}
```

4. Write JavaScript code to move absolutely-positioned car image when a user clicks the move button and stop the car image when a user clicks the stop button on the following webpage.

```
//sample code to move image
function init()
{
imgObj = document.getElementById('img');
imgObj.style.position= 'relative'; imgObj.style.left = '0px';
 }
 function moveRight()
{
imgObj.style.left = parseInt(imgObj.style.left) + 1 + 'px';
animate = setTimeout('moveRight()',10); // call moveRight in 20msec
 }
 function stop()
{
 clearTimeout(animate);
 imgObj.style.left = '0px';
 }
window.onload =init();
```

5. Create an array of objects, where each object describes about a student and his/her properties for the name (a string), regno (a string) and year of birth (a number). Iterate through the array of students and display it on the web page. For each student, print the student name and register no like so: "John's Register Number is UR13CS129". Now use an if/else statement to categorize the students they have date of birth before 2000 and after 2000.

```
// Create an array of objects
```

```
var student = [ { name:'John', r_num:'123', dob:'20-apr-1995' }, { name:'Joseph', r_num:'456', dob:'05-jan-1998' }];
//Iterate through an array
        for (var i = 0; i < Objectname.length; i++) {
                //access property
                Objectname[i].property1;
                        //student[i].name;


        }
```